

Build a model that explains a direction

Learning intention

Explain the insulin response to raised blood glucose using a qualitative, controllable model.

What could reduce a rise in blood glucose?

We will model a mechanism, without medical numbers.

Message, material or store?

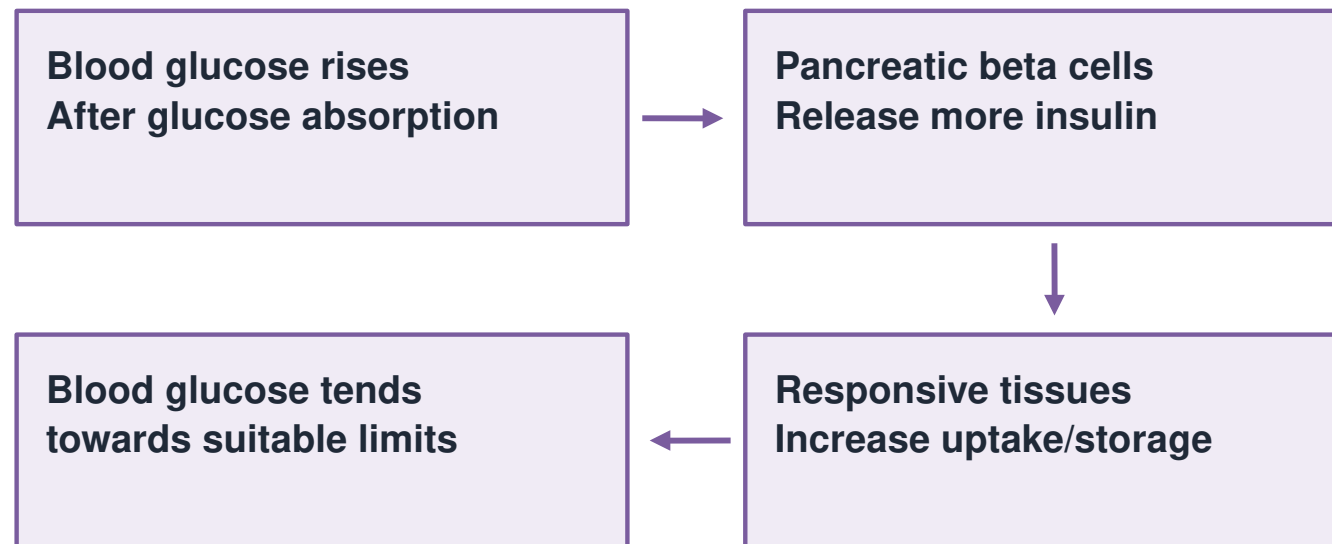
R1. What carries hormones around the body?

R2. What can cells use glucose for?

R3. What does negative feedback oppose?

Trace the core response

Raised glucose triggers a corrective response

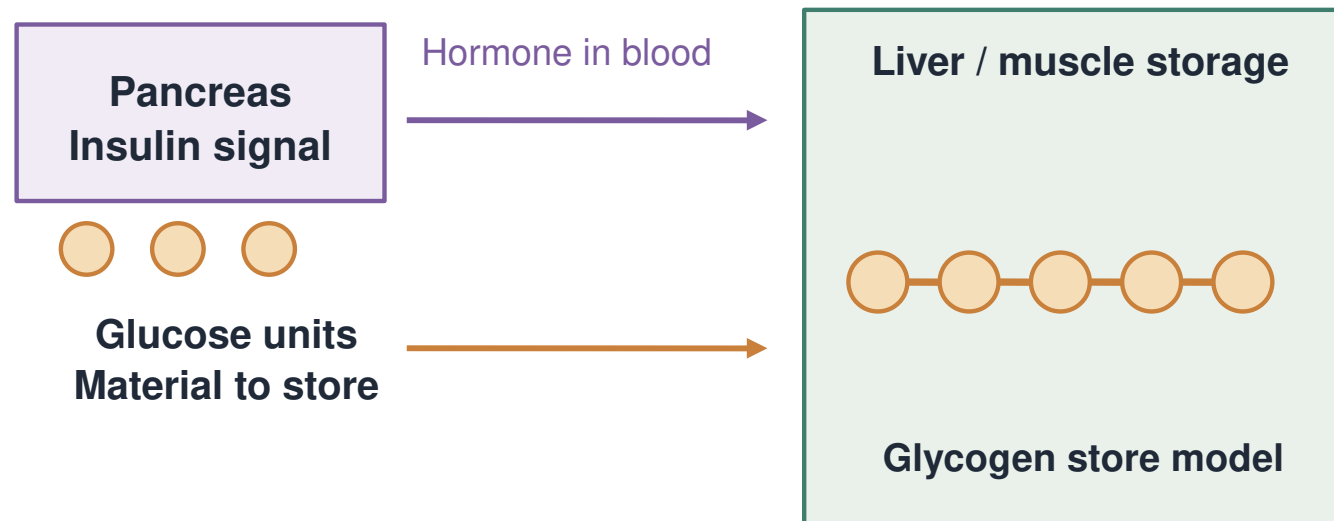


Core model: insulin response to raised glucose.

Raised glucose stimulates pancreatic beta cells. More insulin is released into blood. Responsive tissues change glucose handling.

Follow the material separately from the signal

A signal is not the stored material

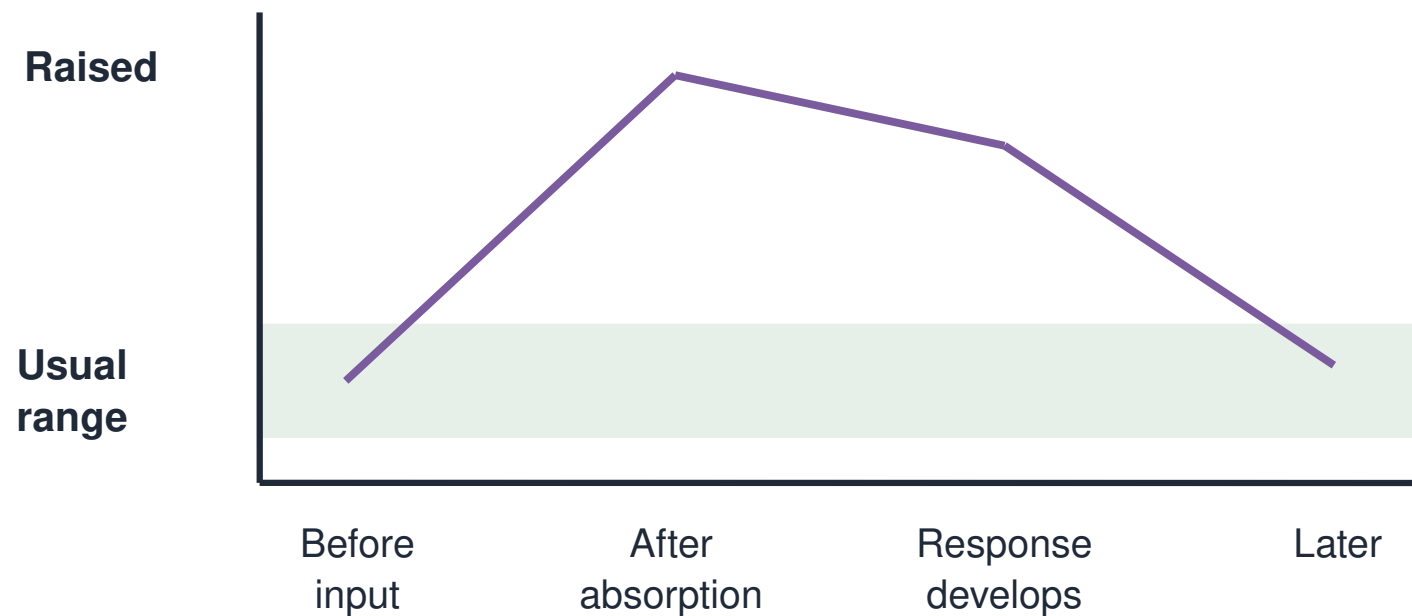


Insulin is a signal; glucose supplies the stored units.

Insulin is a hormone signal. Glucose units can be stored as glycogen. The hormone does not become the store.

A direction model has a clear limit

Read direction, not a clinical number



The trace shows a rise and a corrective response. It has no real time scale, concentration scale or dose rule.

Control-panel challenge

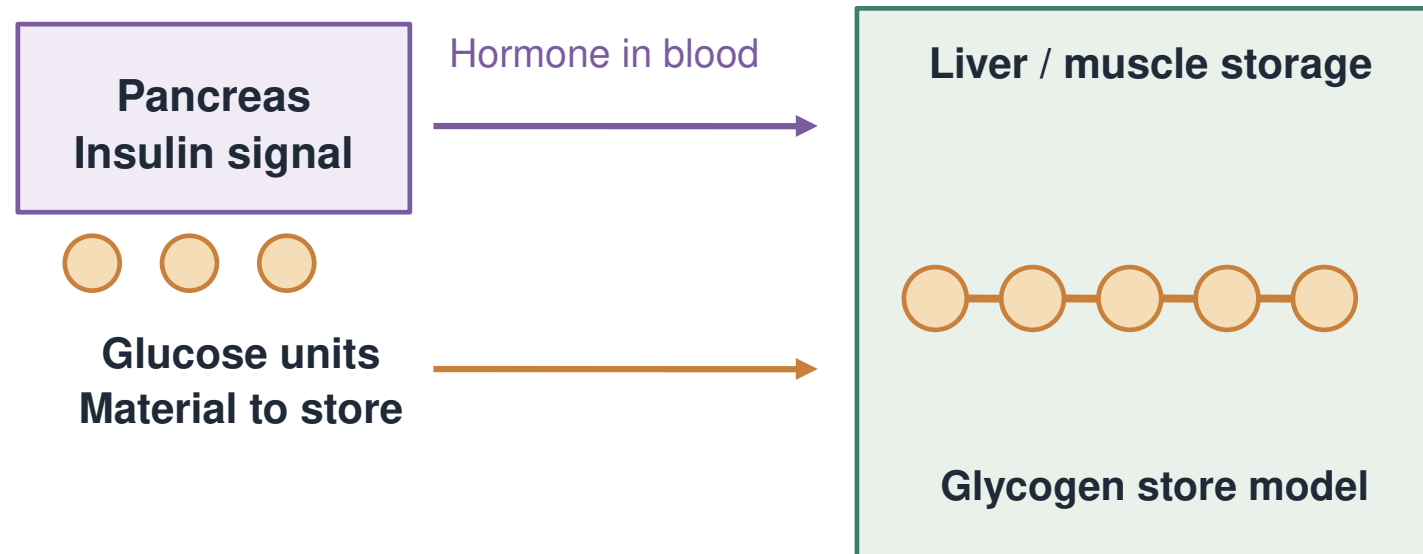
W1. Predict the next stage before advancing the model. Explain the link from glucose rise to insulin release.

Card	Core event
A	Glucose is absorbed into blood; concentration rises.
B	Pancreatic beta cells release more insulin.
C	Insulin-responsive tissues increase glucose uptake/storage processes.
D	The concentration tends towards the usual suitable range; the insulin stimulus reduces.

Qualitative sequence; no individual medical prediction.

Repair the substance swap

A signal is not the stored material

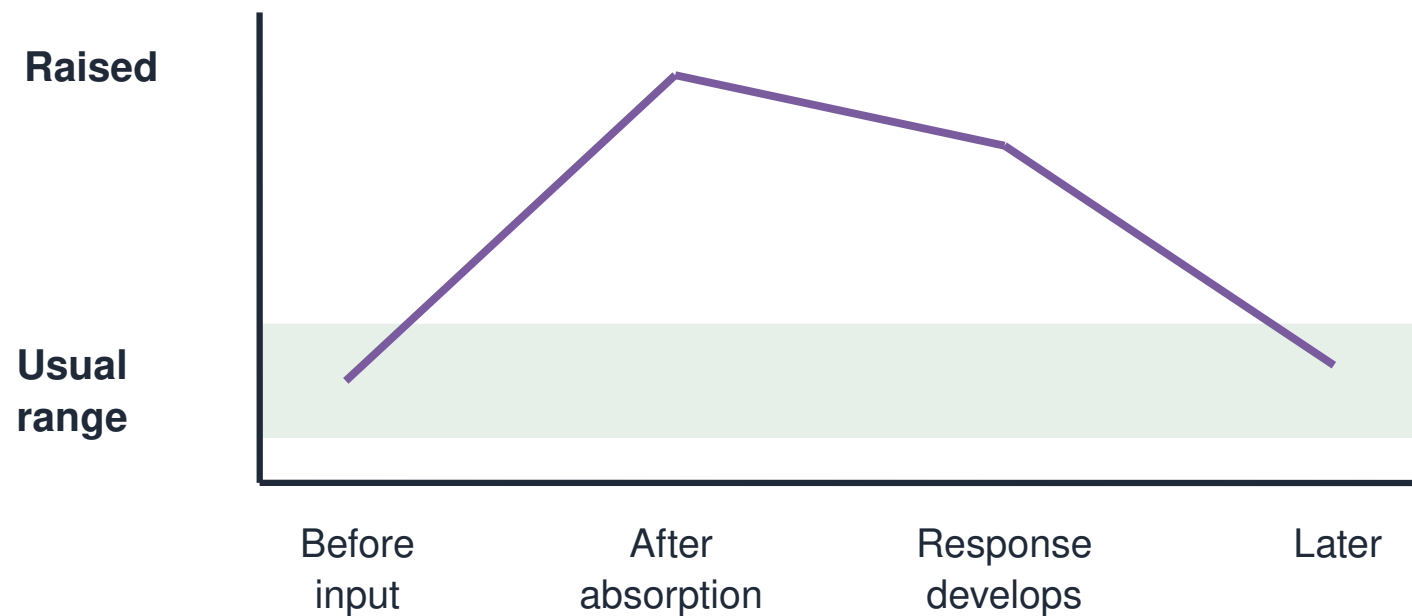


Insulin is a signal; glucose supplies the stored units.

W2. “Insulin is stored as glycogen.” Replace the wrong substance and explain the hormone’s role.

Test what the model cannot claim

Read direction, not a clinical number



W3. Can the trace predict an exact reading after a meal? Name the missing information.

Which response belongs in the core model?

Choose a causal sequence.

A Raised glucose → more insulin release → uptake/storage responses.

B Insulin becomes glucose, so blood glucose rises further.

C The picture supplies an exact medicine dose.

Create the explanation that sits behind the controls

Choose your agreed route.

Core: insulin response. Higher extension: separate glucagon branch.

Predict, explain and state a limit

Use the qualitative sequence and role diagram.

Make one correction to a misleading claim.

Audience: which arrow is the message?

L1. Share one explanation.

Audience asks which substance is signalling and which is stored.

An accurate model is deliberately bounded

It explains a feedback mechanism.

It does not diagnose a condition or prescribe an action.

Keep the roles distinct

E1. Which organ releases insulin?

E2. What is stored as glycogen?

E3. Why can this model not predict a dose?

Next useful step

Sequence • substance roles • feedback • model limits.